Last meeting

Last meeting:

- Presented auction-style model following Hendricks and Porter (1988)
- Discussed some reduced form evidence
- To do: extend model to differentiated products and think about identification

Today's meeting:

- Show model and identification strategy
- Are information asymmetries plausible?

Model Setup

Two-stage game:

- **Stage 1:** Banks compete for customer via introductory product
- **Stage 2:** Home (informed) and non-home (uninformed) compete for a loan

Stage 2 primitives:

- Bank 1 (home) observes default probability $h \sim F(h)$ on $[h_{\min}, h_{\max}]$ and plays $\sigma(h)$
- Bank 2 (non-home) only knows $F(h)$ and plays r_2 for all borrowers
- Consumer utility from bank j:

$$u_{ij} = -\alpha\, r_{ij} - \lambda \cdot \mathbf{1}(j \neq 1) + \varepsilon_{ij}$$

- Market share of incumbent (bank 1):

$$s(r_1, r_2) = \frac{1}{1 + \exp\big(\alpha(r_1 - r_2) - \lambda\big)}$$

Logit Demand and Market Shares (Previous Version)

- Profit functions:

$$\pi_1(r_1; h, r_2) = \big[(1-h)r_1 - 1\big] \cdot s(r_1, r_2)$$

$$\pi_2(r_2; \sigma) = \int_{h_{\min}}^{h_{\max}} \big[(1-h)r_2 - 1\big] \cdot \big[1 - s(\sigma(h), r_2)\big]\, dF(h)$$

- Equilibrium strategies:

$$\sigma^*(h) = \underbrace{\frac{1}{1-h}}_{\text{break-even}} + \underbrace{\frac{1}{\alpha\big(1 - s(\sigma^*(h),\, r_2^*)\big)}}_{\text{markup}}$$

$$r_2^* = \frac{1}{1-\tilde{h}} + \frac{1}{\alpha} \cdot \frac{\int (1-s)(1-h)\, dF}{\int s\,(1-s)(1-h)\, dF}$$

where $\tilde{h} = \frac{\int h\, s(1-s)\, dF}{\int s(1-s)\, dF}$ and all shares evaluated at $(\sigma^*(h), r_2^*)$.

Demand and Equilibrium Prices

- Profit functions:

$$\pi_1(r_1; h, r_2) = \big[(1-h)r_1 - 1\big] \cdot s(r_1, r_2)$$

$$\pi_2(r_2; \sigma) = \int_{h_{\min}}^{h_{\max}} \big[(1-h)r_2 - 1\big] \cdot \big[1 - s(\sigma(h), r_2)\big]\, dF(h)$$

- Equilibrium strategies:

$$\sigma^*(h) = \underbrace{\frac{1}{1-h}}_{\text{break-even}} + \underbrace{m_1(\sigma^*(h), r_2^*)}_{\text{markup}}$$

$$r_2^* = \underbrace{\frac{1}{1-\hat{h}}}_{\text{break-even}} + \underbrace{E_w[m_2]}_{\text{weighted avg. markup}}$$

- $\hat{h} = \frac{\int h\, s_2\, dF}{\int s_2\, dF}$: default prob. of marginal borrower. $E_w[m_2] = \frac{\int m_2\, s_2(1-h)\, dF}{\int s_2(1-h)\, dF}$ is the markups of the marginal borrower adjusted for their repyament probability. where $m_1 \equiv \frac{s}{-s_1}$, $m_2 \equiv \frac{s_2}{-s_2}$ are the inverse semi-elasticities

Equilibrium Characterization

Key properties:

- $\sigma^*(h)$ is strictly increasing in h
- Markup is *decreasing* in h: riskier borrowers are more contested
- Adverse selection: $\bar{h}_1 < E[h] < \bar{h}_2$

Stage 1: Backward Induction

Continuation values from Stage 2:

$$V^I = \int \left[(1-h)\sigma^*(h) - 1\right] \cdot s(\sigma^*(h), r_2^*)\, dF(h) \qquad \text{(incumbent)}$$

$$V^E = \pi_2(r_2^*; \sigma^*) \qquad \text{(entrant)}$$

Net value of acquiring the customer: $\Delta V = V^I - V^E > 0$

Stage 1 pricing: Each bank sets introductory price $p^{(1)}$ solving:

$$\max_{p^{(1)}} \; q(p^{(1)}, p_{-j}^{(1)}) \cdot \left[p^{(1)} - (c^{(1)} - \Delta V)\right]$$

- Effective marginal cost is $c^{(1)} - \Delta V < c^{(1)}$: **loss-leading** on the introductory product
- Higher λ or information asymmetry $\Rightarrow$ higher ΔV $\Rightarrow$ lower introductory prices
- Net welfare effect is ambiguous

Identification: Overview

Observables: winning rate r_i^{win}, winner identity $W_i \in \{1, 2\}$, default $D_i \in \{0, 1\}$

Sequential identification in 4 steps:

Object	Identified from	Step
r_2^*	Winning rate when $W_i = 2$ (constant)	1
$\sigma^*(h)$, $\tau(r)$	$E[D \mid r, W = 1]$ (defaults reveal types)	2
α	Markup curvature across types	3
λ	Markup level given α	3
$F(h)$	Selection-corrected type distribution	4

Identification: Steps 1–2

Step 1: r_2^* is directly observed.

- Bank 2 plays a single rate $\Rightarrow$ all bank-2 winners have the same $r_i^{\text{win}} = r_2^*$

Step 2: $\sigma(h)$ from defaults among bank-1 winners.

- When bank 1 wins: $r_i^{\text{win}} = \sigma^*(h_i)$ and $D_i \sim \text{Bernoulli}(h_i)$
- Conditional default rate reveals the type:

$$E[D_i \mid r_i^{\text{win}} = r,\ W_i = 1] = h_i = \tau(r)$$

- This recovers $\tau = \sigma^{-1}$ **nonparametrically**
- Yields data objects: $h_i = \tau(r_i^{\text{win}})$, markup $m_i = r_i^{\text{win}} - \frac{1}{1-h_i}$, rate gap $\Delta_i = r_i^{\text{win}} - r_2^*$

Identification: Step 3 — α and λ

- From home bank FOC:

$$\ln\big(\alpha\, m(h) - 1\big) = \lambda - \alpha\, \Delta(h)$$

  - Must hold for *every* type h
  - $m(h)$ and $\Delta(h)$ are data; α and λ are unknowns
- Separating α from λ:
  - Evaluate at two types $h_a \neq h_b$ and subtract:

$$\ln \frac{\alpha\, m(h_a) - 1}{\alpha\, m(h_b) - 1} = \alpha \left(\Delta(h_b) - \Delta(h_a)\right)$$

  - Single equation in single unknown α
  - Then recover: $\lambda = \ln(\alpha\, m(h) - 1) + \alpha\, \Delta(h)$

Intuition:

- α = *slope*: how markup responds to rate gap across types
- λ = *level*: uniform shift in markups for all types

Identification: Step 4 — Type Distribution $F(h)$

With α, λ known, market shares are identified:

$$s(h) = \frac{1}{1 + \exp\big(\alpha\,\Delta(h) - \lambda\big)}$$

Selection correction: Among bank-1 winners, the density of types is:

$$f_{h|W=1}(h) = \frac{s(h)\,f(h)}{\Pr(W_i = 1)}$$

Inverting:

$$f(h) = \frac{f_{h|W=1}(h) \cdot \Pr(W_i = 1)}{s(h)}$$

- $f_{h|W=1}$ is estimated from the empirical distribution of $h_i = \tau(r_i^{\text{win}})$ among bank-1 winners
- $s(h) > 0$ for all h (logit) $\Rightarrow$ every type observed among bank-1 winners